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Phosphate affects susceptivity of *Sarcodia suae* to arsenate

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ABSTRACT

Algal susceptivity to inorganic arsenate As(V) is linked to the plant nutrient phosphate due to their similar uptake processes. This study determines the response of *Sarcodia suae* to three phosphate concentrations (1, 10 and 20 mg l^{-1}) when exposed to As(V) (0, 250 µg l^{-1}) for 7 days. The parameters measured include growth, pigments (Chlorophyll *a* (Chl *a*), allophycocyanin (APC), phycocyanin (PC), phycoerythrin (PE), total phycobiliproteins (TPBP) and carotenoids), inorganic arsenic (iAs), and total arsenic (TAs) uptake behaviour and bioaccumulation capacity. The growth rate showed significant differences among treatments. Significantly higher levels of pigments including Chl *a* and carotenoids were observed with an increase in external phosphate concentrations. On the other hand, APC, PC, PE and TPBP showed no significant differences among treatments. The trend of iAs and TAs uptake when phosphate is high suggests competition between phosphate and As(V) as is evident from the reduction of uptake with increased phosphate. The depletion of phosphate and As(V) in the medium also suggests accumulation by the algae. In addition, *S. suae* was considered an As accumulator but not a hyperaccumulator as defined on its bioaccumulation capacity of 48 and 97 mg kg^{-1} of iAs and TAs, respectively. Overall, these findings suggest that environmental factors including levels of phosphate have a key role in developing effective As bioremediation in the environment.

HIGHLIGHTS

- Phosphate in *Sarcodia suae* probably regulates arsenic uptake.
- Photosynthetic pigments can be linked to competitive mechanisms of As(V).
- The red alga *Sarcodia suae* is a candidate for As wastewater treatments by Bioconcentration Factor analysis.

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KEYWORDS Algae; arsenate; culture; growth; phosphate; phosphate uptake; pigment; Rhodophyta; *Sarcodia suiae*; toxicity

Introduction

Arsenic (As) originating from natural and anthropogenic sources has reached high concentrations in natural water systems, ranging from less than 0.5 µg l^{-1} to more than 5000 µg l^{-1} (see Smedley & Kinniburgh, 2002 for a review), and reaching up to 200 000 µg l^{-1} in certain saline lakes in California, USA (Oremland *et al.*, 2004). Arsenic species comprise arsenate As(V), arsenite As(III), monomethylarsonic acid (MMAA), dimethylarsinic acid (DMAA) and other complex arsenic compounds such as arsenobetaine, arsenocholine and arsenosugars (Argese *et al.*, 2005). Among these, pentavalent arsenates that occur in a variety of oxidizing environments are considered a subclass of the phosphate mineral group owing to the similarity in size and charge of the phosphate and arsenate anionic unit. As(V) exists primarily as the deprotonated oxyanion [H_2AsO_4] and is incorporated by macroalgae due to its similarity to the major seawater nutrient phosphate, which also exists primarily as the deprotonated oxyanion [H_2PO_4] (Francesconi, 2010; Sele *et al.*, 2012). As(V) as an analogue of phosphate could enter macroalgae competitively through phosphate transport systems (Zhao *et al.*, 2009). Specifically, arsenate could enter macroalgae through the PST (Phosphate Specific Transport) and PIT (Inorganic Phosphate Transport) systems (Thiel, 1988; Ybarra, 1998). The PST system is slow and 100 times more specific for phosphate than arsenate, while the PIT system is fast and does not discriminate between phosphate and arsenate (Ybarra, 1998). In turn, As (V) could be recognized as a phosphate, an essential nutrient involved in many biochemical pathways (Goyer & Clarkson, 2001; Hughes, 2002) including growth. Algae may take up both As(V) and phosphate through phosphate transporters because they share the same internalization mechanisms (Wang *et al.*, 2013). During the entry process into macroalgae, As(V) substitutes for phosphate leading to the production of unstable arsenical by-products. It interferes with the normal phosphorylation processes

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resulting in uncoupling of oxidative phosphorylation, and thereby disruption of ATP (adenosine 5′-triphosphate) synthesis (Goyer & Clarkson, 2001). As a result, As(V) reacts enzymatically with ADP (adenosine 5′-diphosphate) to form an arsenate–phosphate bond. The arsenical ester formed is unstable and spontaneously hydrolyses to ADP and As(V), thus preventing ATP formation. In the same way, ATP-dependent transport, glycolysis, the pentose phosphate pathway, and signal transduction pathways could be impaired (Slyemi & Bonnefoy, 2012).

It is, therefore, anticipated that As(V) toxicity (i.e. the substitution of phosphate-containing biological molecules such as phospholipids, ADP, ATP and ion pumps) might depend on phosphate availability in the environment and subsequent synthesis of phosphate transporters (Miot *et al.*, 2009). Several studies supported the notion that arsenic toxicity is phosphate condition-dependent so the combined As/Phosphorus ratio determines arsenate toxicity (Wang *et al.*, 2013). Phosphate enrichment reduced arsenic toxicity due to extracellular competition (Levy *et al.*, 2005). On the other hand, some studies do not support the traditional view that phosphate-starved cells are more sensitive to arsenate (Thiel, 1988; Wängberg & Blanck, 1990). Studies by Gong *et al.* (2009*a*) and Knauer & Hemond (2000) showed different results for *Microcystis aeruginosa* and *Chlorella* sp., where arsenate tolerance was found to be unaffected by ambient phosphate concentration and phosphate-limited conditions suggesting that the traditional view of phosphate-starved cells being much more prone to arsenic toxicity may not be correct. Arsenic toxicity was confusing due to contradictory results, and the extent to which phosphate affects uptake patterns in algal species remains unclear.

The marine red algal genus *Sarcodia*, belonging to the family Sarcodiaceae, consists of 15 species that are widely distributed across the world, including in the Indo–Western Pacific region, West Africa, Antarctica, northern New Zealand, and Taiwan (Guiry & Guiry, 2021). *Sarcodia suae* is common in the coastal and intertidal areas of Taiwan and has been confirmed to have high antioxidant activity and rich dietary fibre content, displaying potential for mass production as a food source or as an aquatic feed, or for use in cosmetics and cosmeceuticals (Wu *et al.*, 2019; Lee *et al.*, 2020). We have investigated the cultivation and physiological analysis of *S. suae* since it was first collected in an intertidal zone in Wan-li-tong, Pingtung, south Taiwan in 2014. Biochemical composition of the wild-type alga was 16.6% crude protein, 0.12% crude lipid, 2.86% crude fibre, 40.36% nitrogen-free extract (mostly carbohydrates), 30.25% ash and 9.54% moisture (% of dry weight). Carbohydrates were predominant. Previous studies have also demonstrated that one of the main roles that peptidoglycans, lipopolysaccharide, and insoluble carbohydrates play in the cell wall is to attach strongly to heavy metals through active enriched electronegative groups (Moreno-Sánchez *et al.*, 2017). Therefore, we considered *S. suae* to be an ideal model species for the exploration of these toxicological mechanisms, given its biochemical composition and its history of sustainable artificial cultivation by man.

While the issue of arsenate accumulation in macroalgae is important, few references in the literature involving studies in indoor systems can be found. We put this down to the difficulty of finding suitable macroalgae that can be cultured in indoor systems for a prolonged period of time. This study uses an indoor culture system to investigate the association of As(V) and phosphate with As biochemistry, to clarify the effect of arsenate and phosphate on algal growth and pigment composition (Chlorophyll *a* (Chl *a*), carotenoids, allophycocyanin (APC), phycocyanin (PC), phycoerythrin (PE) and total phycobiliproteins (TPBP)), and to establish the existence of a competitive relationship between arsenate and phosphate accumulation in the red alga *S. suae*. This should be useful in establishing an experimental method for assessing the accumulation of arsenate in macroalgae, an important step for exploring the likelihood of arsenate accumulation with applications in sewage disposal and food safety.

Materials and methods

Algal collection, maintenance and pre-culture

The marine red alga *Sarcodia suae* S.-M.Lin & C. Rodríguez-Prieto was collected in July 2014 in an intertidal zone in Wan-li-tong, Pingtung, south Taiwan, (21°59'41"N, 120°42'22"E). The samples were placed in a ventilated breathable box and immediately transported to the laboratory. The algae were then washed with 1.5% povidone-iodine and 2 µ-filtered UV-irradiated sterilized seawater to remove any adhering debris or epiphytic organisms, and were maintained in a 1 t fibreglass tank. The tank was supplemented with air and continuously filtered seawater (30 l day^{-1}) under natural sunlight. Before the start of the experiment, healthy adult *S. suae* algae with a wet weight of 7.3 ± 0.5 g were selected and pre-cultured at 25 ± 1°C (temperature maintained by thermostat equipment; SS-980, Tominaga, Taipei City, Taiwan) under a photoperiod of 12:12 h (light: dark) and an irradiance of 80 ± 10 µmol photons m^{-2}

s^{-1} for 2 weeks in sterile PES medium in a 20 l clear plastic bottle, in July 2019. All the materials used in the culturing process were autoclaved (121°C for 20 min at 15–20 psi) and precautions were taken to ensure that the cultures were free from microbial contamination.

Phosphate and arsenate treatments

In order to investigate short-term variation of both phosphate and arsenate concentration in *S. suae* in this study, PES culture medium was added on the initial day of the experiment to maintain the viability of the algae for the entire experiment period. Based on our previous observations, *S. suae* were in good condition after 7 days of cultivation. After that, following the decrease in nutrition concentration, the growth of *S. suae* gradually declined. To ensure that arsenate uptake was only affected by phosphate and arsenate, we provided full nutrition for the algae within the experimental period (7 days) to investigate the accumulation of both phosphate and arsenate after 7 days cultivation. The specific process is described below. The culture medium was prepared in a 1 l beaker containing sterilized seawater enriched with PES medium. Each beaker contained an average weight of 7.3 ± 0.5 g *S. suae*. The culture medium was enriched with a range of phosphate concentrations (1, 10 and 20 mg l^{-1}) under 250 µg l^{-1} As(V) for 7 days of the culture period. The As(V) species used was sodium arsenate dibasic heptahydrate (As(V); $Na_2HAsO_4.7H_2O$, purity 98.0%, Sigma-Aldrich, USA) and the phosphate source was potassium phosphate dibasic (K_2HPO_4; reagent grade ≥ 98.0%, Sigma-Aldrich, USA). Phosphate concentrations were designed based on previous studies on arsenic and phosphate interactions for macro- and micro-algae (Duncan *et al.*, 2013; Bahar *et al.*, 2016; Baker & Wallschläger, 2016; Lee *et al.*, 2019; Mamun *et al.*, 2019*a*). The control groups had no added As(V) and/or phosphate (natural seawater). A total of four treatments with three replicates were designed (Table 1). The cultures were maintained at 25°C in a specialized incubator with a temperature controller (SS-980, Tominaga, Taipei city, Taiwan). The light was supplemented using artificial light from a fluorescent bulb with an irradiance of µmol photons m^{-2} s^{-1} in a 12:12 light/dark photoperiod. Light intensity was measured using a Lighting Passport device (Asensetek Incorporation, New Taipei city, Taiwan). The culture media were not changed until the end of the experiment. To avoid the loss of arsenate and phosphate, water samples were only collected for phosphate and arsenic analysis on days 0 and 7 of the culture period. Phosphate analysis was conducted directly on the isolated samples, while arsenic analysis involved the addition of 1% 1 M HCl.

Table 1. Initial concentrations of As(V) and P into the culture medium of *Sarcodia suae*.

Abbreviations	Phosphate (mg l^{-1})	As(V) (µg l^{-1})
P0 + $As^5$0 (natural seawater)	0	0
P1 + $As^5$250	1	250
P10 + $As^5$250	10	250
P20 + $As^5$250	20	250

Measurement of growth

Growth measurements were determined by taking the fresh weight of each algal sample before and after the experiment. The average fresh weight and growth rate was calculated based on Mamun *et al.* (2019*b*).

$$GR = \left[\frac{ln\left(\frac{Wt}{W0}\right)}{t}\right] \times 100$$

where W_t is the fresh weight (g) of the algae at the end of each exposure, W_0 is the fresh weight (g) of the algae at the beginning of the experiment, and t is the culture time in days.

Measurement of pigments

The extractions of pigments (APC, PC, PE, TPBP, Chl *a* and carotenoids) were based on the procedures of Bennett & Bogorad (1973). Briefly, 10 pieces of the ceramic sphere (Wheaton, Illinois, USA), 0.3 g of Garnet Matrix A bulk Powder (MP Biomedicals, California, USA), and a freeze-dried sample of *S. suae* (0.05 g) were weighed in 2 ml Eppendorf tubes. About 1 ml of PBS (Phosphate Buffered Saline) was added and centrifuged using FastPrep-24 5G (MP Biomedicals, California, USA) at a speed of 7 min for 20s and held for 10–15 s. The samples were centrifuged and the supernatant was collected to be utilized for pigment quantification. The pellet was retained and the procedure repeated thrice. Likewise, Chl *a* and carotenoids were measured by adding 5 ml of 90% acetone to the pellet in a glass test tube capped with cork stopper, kept dark overnight. The pigments were transferred to a cuvette and the absorbance was read in a spectrophotometer and calculated using the following formula based on Bennett & Bogorad (1971) and Vimala & Poonghuzhali (2015).

$$(\text{Phycocyanin, PC})\ \left(\text{mg ml}^{-1}\right) = \frac{OD615 - 0.474\ (OD652)}{5.34}$$

$$(\text{Allophycocyanin, APC})\ \left(\text{mgml}^{-1}\right) = \frac{OD652 - 0.208\ (OD615)}{5.09}$$

$$(\text{Phycoerythrin, PE})\ \left(\text{mg ml}^{-1}\right) = \frac{OD562 - 2.41(PC) - 0.849(APC)}{9.62}$$

$$(\text{Total Phycobilliprotein, TPBP})\left(\text{mg ml}^{-1}\right) = PC + APC + PE$$

$$(\text{Chlorophyll } a, \text{Chl } a)\ \left(\text{mg ml}^{-1}\right) = \frac{(11.64 \times \text{OD663} - 2.16 \times\ \text{OD645} + 0.1 \times \text{OD630}) - 0.0084}{0.0074 \times 60}$$

$$(\text{Carotenoids})\ \left(\text{mgml}^{-1}\right) = OD480 + (0.114 \times OD663) - (0.638 \times OD645)$$

where, OD = absorbance at respective wavelength and APC, PC, PE, TPBP, Chl *a* and carotenoids are concentrations of allophycocyanin, phycocyanin, phycoerythrin, total phycobiliproteins, Chl *a* and carotenoids respectively.

Arsenic bioaccumulation capacity of *Sarcodia suae*

Sarcodia suae were individually exposed to low phosphate (0.3 mg l^{-1}) and 250 µg l^{-1} As(V) in a 1 l beaker over a 7-day culture period. After exposure, the algae were collected and washed three times with distilled seawater to remove externally bound arsenic. The algae were freeze-dried and the arsenic content was determined prior to Bioconcentration Factor (BCF) value determination. BCF expresses the uptake of elements by organisms relative to their availability in the water sample. It is calculated as the ratio of arsenic concentration in algal biomass (dry weight) to the initial arsenic concentration in the culture solution as described by Zayed *et al.* (1998).

$$\text{BCF} = \frac{Cc}{Cw}$$

where Cc (As in algal samplings) expressed in µg g^{-1} dry weight and Cw (initial As concentration in water) in µg l^{-1}.

Sample preparation before arsenic analysis

After exposure for 7 days, *S. suae* samples were collected, washed with deionized fresh water three times, kept in Ziploc freezer bags frozen at −80°C for 24 h, and freeze-dried for 70 h. These samples were powdered by crushing manually, labelled and stored in a Ziploc bag at −20°C until analysis.

Arsenic analyses

The As speciation procedure was performed according to procedures of Khan *et al.* (2015), using High Performance Liquid Chromatography-Inductively coupled plasma mass spectrometry (HPLC-ICP-MS) (X-series, Thermo-Fisher, Waltham, USA) for As speciation. Briefly, powdered freeze-dried *S. suae* weighed into 50 ml self-standing polypropylene volumetric tubes, with ~0.15 g algae used for the arsenic analysis. After weighing, 50% methanol solvent in 1% HNO_3 was added. With loosely sealed caps, the tubes were heated in a heat block system and maintained at 90°C for 30 min, and then at 65°C for 3 h and 30 min. After 30 min of cooling, the tubes were washed with double distilled water and transferred to Falcon tubes to a final volume of 50 ml. The supernatant was collected and filtered through 0.45 µm pore-size nylon membrane filters. After filtration, the samples were diluted prior to the quantification of the resulting solution. Arsenic species were separated and quantified by an anion exchange column (Hamilton PRP-X100, 10 µm in diameter and 4.6 mm × 250 mm in diameter) with a mobile phase of 2 mM ammonium bicarbonate in 1% methanol (MeOH), pH 8.0 (Mobile Phase A) and 20 mM ammonium nitrate and 20 mM ammonium phosphate in 1% MeOH, pH 9.2 (Mobile Phase B).

The TAs content was measured following the method of Ackley *et al.* (1999) and Xue *et al.* (2012) with modifications. A 0.15 g crushed sample in 10 ml water was weighed and digested with 2 ml of 65% concentrated nitric acid (Guaranteed Reagent Grade, Merck, Germany), and heated at 95°C on a hot plate until completely dissolved. After cooling to room temperature, the samples were transferred into Falcon tubes, and Millipore ultrapure water was added to increase the volume to 50 ml. The samples were then filtered with a 0.22 µm disposable nylon syringe filter for analysis. The TAs content was determined using ICP-MS (X-series, Thermo-Fisher, Waltham, Massac-husetts, USA).

Quality assurance and quality control for the instruments were checked and validated by measuring several quality parameters including specificity,

linearity, precision, accuracy and quantitative limits. Reagent blanks and Certified Reference Materials (NMIJ CRM 7405-a Hijiki Seaweed) (National Research & Development Corporation Industrial Technology Research Institute, Japan) were included to assure the accuracy of the test method and are within 100 ± 20% of the expected quality control checks. The operating conditions of HPLC and ICP-MS including measurement parameters were followed as described by Khan *et al.* (2015) with 100 ml as the injection volume of the samples during analysis. All solutions used for the ICP-MS were of analytical reagent grade. The minimum detectable concentrations were 0.02 and 0.05 for TAs and iAs, respectively.

Phosphate analysis

P content in the culture medium was determined in reference to standards (US Environmental Protection Agency, 1983; American Public Health Association, 1998), using water samples of ~50 ml collected from day 0 and day 7.

Data analysis

One-way analysis of variance (ANOVA) with a subsequent Duncan Multiple Range Test was used to identify significant differences between the effects of different phosphate levels and As(V) for various factors, i.e. growth rate, pigments (Chl *a*, carotenoids, APC, PC, PE and TPBP), and iAs and TAs uptake in the algal body and in the medium, as well as phosphate levels in the culture medium. The significance level applied to all statistical analyses was set at $p < 0.05$, and the results are presented as means ± standard deviation (SD), n = 3. Statistical analyses were carried out using SAS (Statistical Analysis System) Software Version 9.0 (SAS Institute, USA).

Results

Effects of phosphate and As(V) on growth attributes of Sarcodia suae

The fresh weight of algae in all treatments at 7 days shows an increase over their initial weight (Fig. 1). This suggests that the different cultural conditions were adaptable by *S. suae* that led to the increase in biomass of the algae.

In terms of specific growth rate, the highest growth rate (2.23%) was found in the culture condition enriched with high phosphate (P20 + $As^5$250), while a lower growth rate was found in culture conditions enriched with lower phosphate (P10 + $As^5$250 at 1.49% and P1 + $As^5$250 at 0.92%), with the control groups (1.34%) having no significant differences (Fig. 1). The noted maximum growth rate in high phosphate culture conditions is likely a

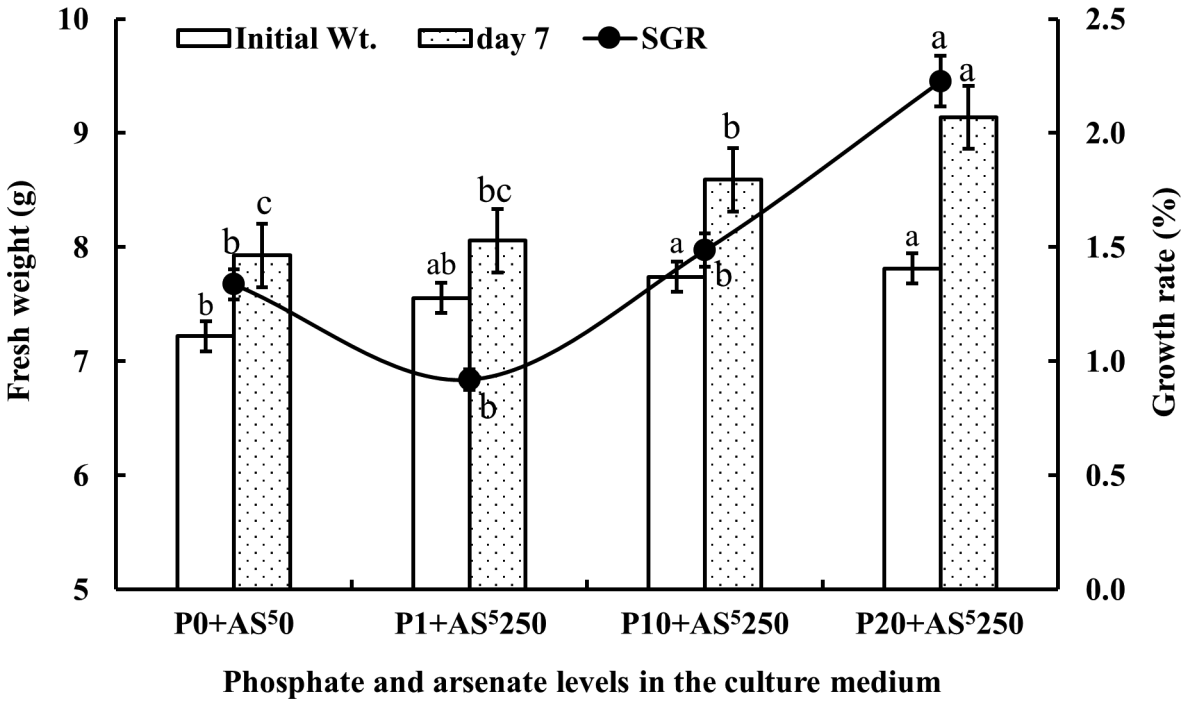


Fig. 1. Effects of different P levels (0, 1, 10, and 20 mg l^{-1}) exposed to 250 µg l^{-1} of As(V) on the changes of fresh weight and growth rate (GR) of *Sarcodia suae* after 7 days. P0, P1, P10 and P20 denote 0, 1, 10 and 20 mg l^{-1} phosphate. $As^5$0 and $As^5$250 denote 0 and 250 µg l^{-1} arsenate. Results represent mean ± SD of three replicates. Different letters indicate significant differences among culture conditions (DMRT, $p \leq 0.05$).

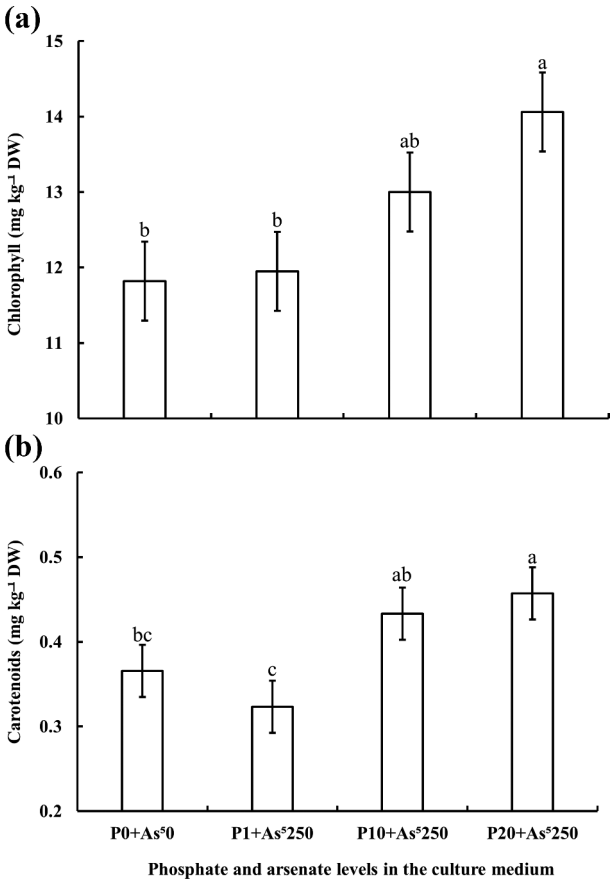


Fig. 2. Responses of (A) Chl *a* and (B) carotenoids in *Sarcodia suae* to P levels (0, 1, 10 and 20 mg l^{-1}) and As (V) (250 µg l^{-1}) after 7 days culture period. P0, P1, P10 and P20 denote 0, 1, 10 and 20 mg l^{-1} phosphate. $As^5 0$ and $As^5 250$ denote 0 and 250 µg l^{-1} arsenate. Results represent mean ± SD of three replicates. Different letters indicate significant differences among culture conditions (DMRT, $p \leq 0.05$).

positive effect of phosphate where phosphate was effectively used for nutrition.

Effects on pigment synthesis of Sarcodia suae

In this study, Chl *a* contents were significantly different among treatments (Fig. 2A). The exposure of the *S. suae* overall resulted in significant increase of Chl *a* with increasing concentrations of phosphate. Similarly, carotenoid contents were observed to be highest at the highest supplementation of phosphate. This implies that various concentrations of phosphate affect the pigment contents in *S. suaie*. On the other hand, photosynthetic pigments from the light-harvesting phycobiliprotein along with APC, PC, PE and TPBP which were an accessory pigment to chlorophyll showed no significant differences among treatments (Fig. 3A–D).

Bioconcentration factor of Sarcodia suae

Generally, organisms can further be recognized based on their index of accumulation potential calculated from their bioconcentration factor (BCF) value. According to Islam *et al.* (2013), an organism with a BCF of less than 1 is considered to be an excluder. An organism exhibiting any metal uptake with BCF of 1 to less than 1000 is considered an accumulator, while a hyperaccumulator were those organisms having a BCF greater than 1000. In our study, the calculated BCF for TAs was 96.56 mg kg^{-1} dry weight while iAs obtained a BCF value of 47.96 mg kg^{-1} dry weight (Table 2). These findings demonstrated the potential of *S. suae* to absorbed As.

Effects of phosphate in total and IAs as accumulation by Sarcodia suae

In this study, there was a significant difference ($p < 0.05$) in the bioaccumulation capacity of TAs and iAs by *S. suae* among treatments (Fig. 4A–B). As expected, minimum or no TAs and iAs were detected when algae were treated with P0 + $As^5 0$ (Fig. 4A–B). On the other hand, trends of TAs and iAs in algae were similar in treatments with phosphate regimes, respectively. As shown in Fig. 4A–B, relatively higher amounts of As were detected in the algal body with low phosphate (P1 + $As^5 250$) with ~51.97 and 39.86 mg kg^{-1} of TAs and iAs, respectively (Fig. 4A–B). Treatments in P10 + $As^5 250$ showed comparatively lower amounts of TAs and iAs (43.44 and 22.22 mg kg^{-1}) compared with P1 + $As^5 250$, suggesting lower phosphate (P1 + $As^5 250$) enhanced the uptake of As in *S. suae* (Fig. 4A–B).

iAs and total As in the culture medium

The TAs and iAs were determined in the culture medium in all the treatments in 7 days after exposure. The amount of TAs remaining in the solution showed no significant differences in treatments enriched with phosphate and ranged from 0.29–0.3 mg l^{-1} (Fig. 4C). Whereas, iAs was detected in the culture medium and showed significant differences in treatments enriched with phosphate. The ranges of remaining arsenic were ~0.147–0.253 mg l^{-1}, significantly higher values being found in P10 + $As^5 250$ and P20 + $As^5 250$ treatments (Fig. 4D). Moreover, TAs or iAs in treatments with no arsenate or phosphate were undetected (Fig. 4C–D).

Levels of phosphate in the culture medium

To ascertain the phosphate levels in the culture medium, initial phosphate was investigated at 0

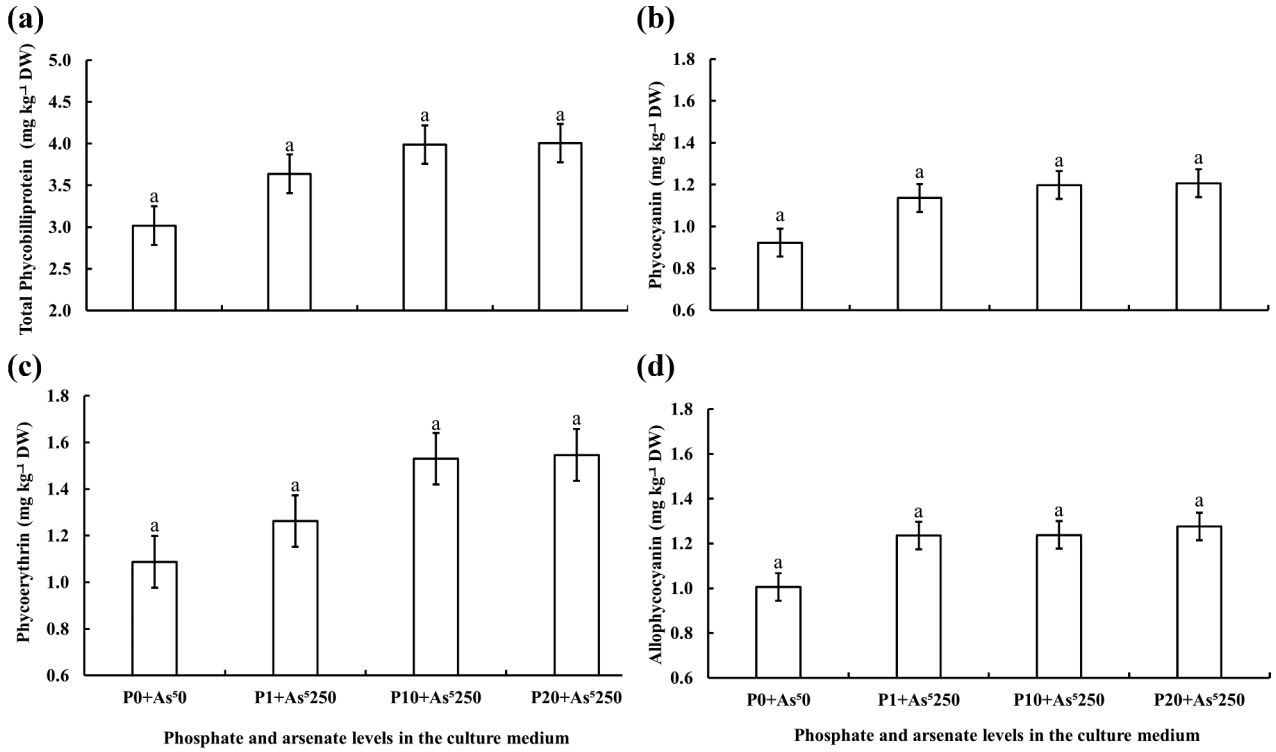


Fig. 3. Responses of (A) total phycobiliprotein, (B) phycocyanin, (C) phycoerythrin and (D) allophycocyanin in *Sarcodia suae* to P levels (0, 1, 10 and 20 mg l^{-1}) and As(V) (250 µg l^{-1}) after 7 days culture period. P0, P1, P10 and P20 denote 0, 1, 10 and 20 mg l^{-1} phosphate. $As^5$0 and $As^5$250 denote 0 and 250 µg l^{-1} arsenate. Results represent mean ± SD of three replicates. Different letters indicate significant differences among culture conditions (DMRT, $p \leq 0.05$).

Table 2. Remediation capacity of *Sarcodia suae*.

Arsenic in algal body (mg kg^{-1} DW)	Accumulation	BCF (mg kg^{-1} DW)
iAs	11.99	48
TAs	24.14	96

days of exposure. In all treatments, the levels of phosphate were in accordance with our phosphate treatments. However, after 7 days of exposure, a significant variation on phosphate levels was found depending on their levels (Fig. 5). The initial concentrations of phosphate at 0, 1, 10 and 20 mg l^{-1} reached 0.000, 0.300, 0.290 and 0.290 mg l^{-1} TAs and 0.000, 0.147, 0.253 and 0.237 mg l^{-1} iAs after 7 days, respectively.

Discussion

Growth parameters

Aquatic organisms such as macroalgae that serve as essential primary producers in the ecosystem are also vulnerable to stressors such as metal and heavy metal stress. The study of the combined effects of As(V) and phosphate on aquatic organisms may provide important information for understanding the risk of environmental concern including arsenic cycling. Growth is a good criterion to determine the response of organisms to toxic compounds since it reflects an organism's metabolism. Sorsa (1983) found that As could inhibit growth, photosynthesis and ATP formation of algae. In the present study, the combined effects of As(V) and phosphate on the growth of *S. suae* showed significant differences among treatments where considerable increase of fresh weight over their initial weight was reflected (Fig. 1).

In terms of growth rate, we found that culture conditions with increasing levels of phosphate generally resulted in higher GR. In fact, it is evident that when phosphate is raised with ranges from 10 to 20 mg l^{-1}, the GR trends were also increasing (Fig. 1) suggesting that phosphate inhibits the toxicity of As (V) and the tolerance was enhanced. This result is also observed in studies on *Microcystis aeruginosa* by Yan *et al.* (2015) and Mamun *et al.* (2019*b*) on *Undaria pinnatifida*. As opposed to the growth rate of control groups and culture medium enriched with low phosphate (P1 + $As^5$250), slightly lower GR (Fig. 1) were observed which could be related to the arsenic accumulated by the algal tissues. Accumulation in algal tissues by the algae in both TAs and iAs in culture medium with minimal phosphate increased the arsenic accumulation, which seems to be connected with the optimal phosphate content for *S. suae* for necessary metabolism for arsenic absorption (i.e. photosynthesis and respiration) (Mamun *et al.*, 2019*a*). It is therefore worthy

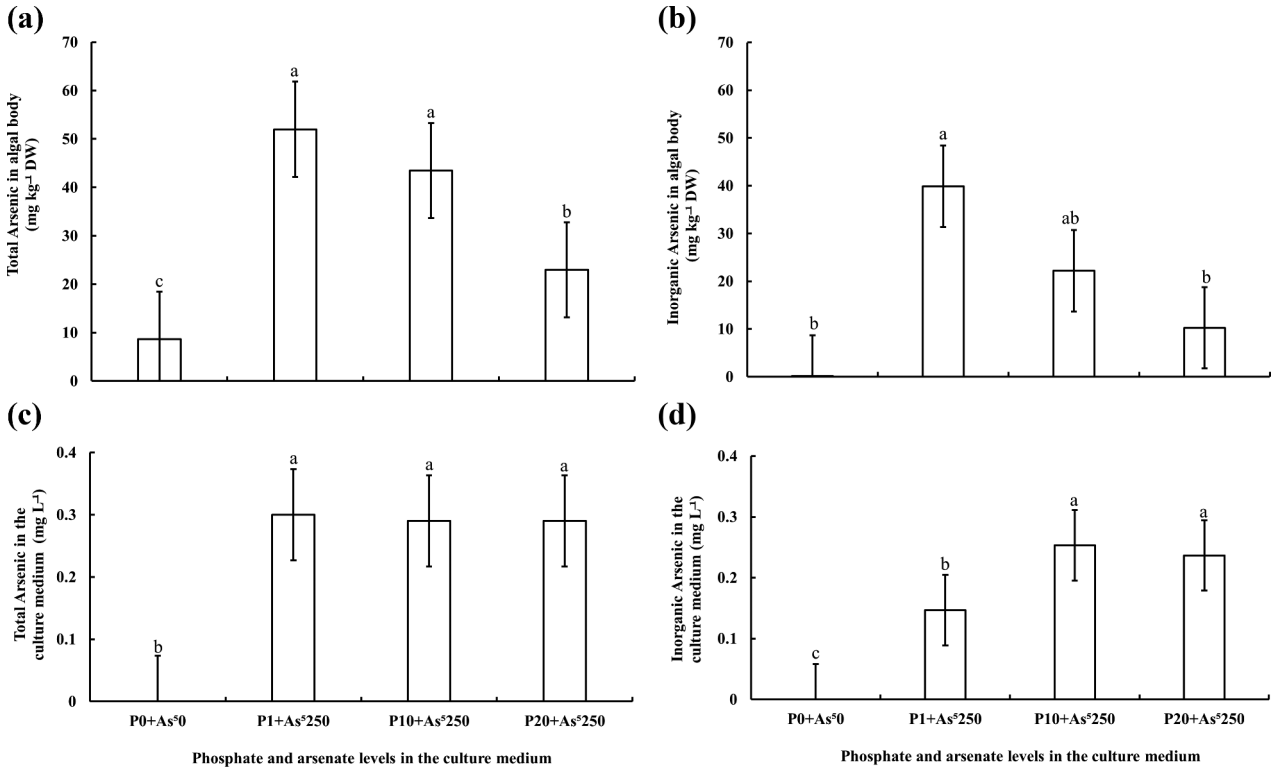


Fig. 4. Total arsenic and inorganic arsenic accumulation into (A–B) algal body and (C–D) culture medium of *Sarcodia suae* to P levels (0, 1, 10, and 20 mg l^{-1}) and As(V) (250 µg l^{-1}) after 7 days culture period. P0, P1, P10 and P20 denote 0, 1, 10 and 20 mg l^{-1} phosphate. $As^5$0 and $As^5$250 denote 0 and 250 µg l^{-1} arsenate. Results represent mean ± SD of three replicates. Different letters indicate significant differences among culture conditions (DMRT, $p \leq 0.05$).

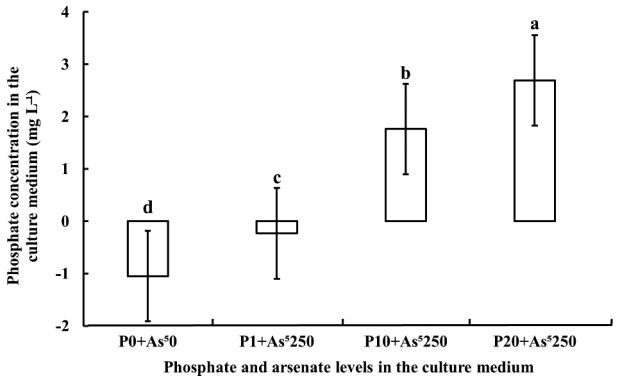


Fig. 5. Phosphate content into the culture medium of *Sarcodia suae* to P levels (0, 1, 10 and 20 mg l^{-1}) and As(V) (250 µg l^{-1}) in 0 days culture period. P0, P1, P10 and P20 denote 0, 1, 10 and 20 mg l^{-1} phosphate. As^{5+}0 and As^{5+}250 denote 0 and 250 µg l^{-1} arsenate. Results represent mean ± SD of three replicates. Different letters indicate significant differences among culture conditions (DMRT, $p \leq 0.05$).

to assume that phosphate signifies importance in arsenic toxicity where a marked decline in growth is related to the effective arsenic accumulation in this culture conditions.

Photosynthetic pigments

Plants' responses, when challenged with stressors such as As, can be characterized according to their biochemical responses. Being a toxic metalloid, arsenic is likely to cause disturbances in plant metabolism and is very likely to affect the photosynthetic machinery of plants (Srivastava & Sharma, 2014). Algae that thrive in the aquatic system are the most vulnerable group of organisms to iAs toxicity (Rahman *et al.*, 2012). As with As(V) oxidation state is absorbed by algae through phosphate transporters, owing to its chemical similarities with phosphate (H_2PO_4 versus H_2AsO_4) (Taylor & Jackson, 2016). Since phosphate is a chemical analogue of As, it may compete with As during uptake (Meharg & Macnair, 1992). Therefore, the enrichment of phosphate in the medium has been suggested to reduce arsenic toxicity because of the competition between them.

Several studies found some plants to be susceptible to As(V) and become more resistant by raising the level of intracellular phosphate (Geng *et al.*, 2006; Wang & Duan, 2009). Others also found that the growth and pigment content (i.e. Chl *a*) of *Microcystis aeruginosa* exposed to As(V) ranging from 108 to 104 mol l^{-1} showed no difference from that of the control (Gong *et al.*, 2009*a*, 2009*b*). Similarly, studies in *M. aeruginosa* in Taiwan showed that phosphate-starved conditions being more sensitive to arsenate toxicity may not be necessarily correct in which algal tolerance was not affected by extracellular phosphate content (Gong *et al.*, 2009*a*). In this study, it was noted from the results that Chl *a* content was increased with the increasing

concentrations of phosphate and a more pronounced increasing trend was significantly observed in P20 + $As^5$250 (Fig. 2A). Besides, carotenoid content in *S. suae* was also manifested higher in treatments with high supplementation of phosphate (P20 + $As^5$250) when compared among treatments (Fig. 2B). We assumed that significant differences in the findings were due to the significant effects of phosphate levels in the medium.

Despite insignificant synthesis in some of the pigments (APC, PC, PE and TPBP) (Fig. 3A–D), the significant synthesis of Chl *a* and carotenoids by the alga was increased with increasing phosphate supplementation suggesting phosphate levels may reduce the arsenic toxicity in algae. As a result, damage of pigments was inhibited and high yield was maintained. On the other hand, low-phosphate enrichment obtained lower contents. These declines may also be related with the As absorption by the algae which was enhanced by the stress response caused by As(V). Since phosphate is an analogue of As(V), it may possible that its main toxicity results from its interference with the metabolism of phosphorus. Upon entering the macroalgae, As(V) substitutes for phosphate leading to the production of unstable arsenical by-products. It interferes with the normal phosphorylation processes resulting in uncoupling of oxidative phosphorylation, and thereby disruption of ATP (adenosine 5′-triphosphate) synthesis (Goyer & Clarkson, 2001). As(V) then reacts enzymatically with ADP (adenosine 5′-diphosphate) to form an arsenate-phosphate bond. The arsenical ester formed is unstable and spontaneously hydrolyses to ADP and As(V), thus preventing ATP formation. In the same way, ATP-dependent transport, glycolysis, the pentose phosphate pathway and signal transduction pathways could be impaired (Slyemi & Bonnefoy, 2012).

Generally, inhibited biosynthesis of Chl *a* and carotenoids and reduced phosphorylation are the most frequently observed symptoms of metal toxicity (John, 1996; Poskuta *et al.*, 1996; Majeti, 2004). Moreover, a sharp decline in PC content due to As(V) and As(III) stress may be due to lysis of the cell wall and disruption of the thylakoid membrane (Srivastava *et al.*, 2009). Altogether, arsenic toxicity is related to phosphate availability and phosphate-supplemented medium reduced the toxicity. On the other hand, it was noted that phosphate and As(V) starved algae were more sensitive to toxicity based on the result of pigment synthesis which was hampered. It is, therefore, assumed that *S. suae* is well adapted to the arsenic and phosphate concentrations being tested in this study supported by prominent modulation of the photosynthetic machinery.

Bioaccumulation capacity

Arsenic accumulation by algae was different according to species, as well as with different arsenic concentrations. Previous studies in *Euglena* sp. showed an accumulation of 14–315 mg kg^{-1} of As (corresponding to 0.014–0.315 mg g^{-1} dry weight) when exposed to 10–200 mg l^{-1} of As(III) for 10 days (Miot *et al.*, 2008). Tahira *et al.* (2019) revealed an accumulation of 0.27 mg g^{-1} of As(III) in *E. gracilis* when exposed to 0.1 mg l^{-1} AsIII) for 7 days. In the present study, the total arsenic accumulation by the algae was 24.14 mg kg^{-1} when exposed to 0.25 mg l^{-1} As(V) (Table 2). On the other hand, 11.99 mg kg^{-1} iAs were accumulated when exposed to similar As(V) concentrations (Table 2). Furthermore, comparing with other similar Rhodophyta assessed for their As uptake capacity in aquatic environment, the following showed higher total arsenic accumulation than our results: *Porphyra* spp. (50 ± 11.73 mg kg^{-1}), *P. umbilicalis* (28.9–49.5 mg kg^{-1}), *P. crispata* (30.1 ± 1.3 mg kg^{-1}), *Gracilaria gracilis* (32 ± 1 mg kg^{-1}) and *Pyropia endiviifolia* (26 ± 0.5 mg kg^{-1}) but our results were neither lower or higher than *Iridaea cordata* (28 ± 6 mg kg^{-1}) and *Chondrus crispus* (23.2–25.5 mg kg^{-1}) (Ma *et al.*, 2018). Some species in the Chlorophyta showed lower accumulation compared with our results, including *Ulva prolifera* (14.65 mg kg^{-1}), *Gayralia oxysperma* (12.68 mg kg^{-1}) and *U. lactuca* (10.33 ± 3.78 mg kg^{-1}) (Ma *et al.*, 2018). Regarding inorganic arsenic accumulation, our result is comparatively higher compared with similar Rhodophyta as *Palmaria palmata* (1.9 mg kg^{-1}) and *P. umbilicalis* (3 mg kg^{-1}) as well as with those of Chlorophyta, including *Ulva rigida* (0.40 ± 0.29 mg kg^{-1}), *Enteromorpha* spp. (0.37 ± 0.07 mg kg^{-1}) and *U. pertusa* (0.36 ± 0.06 mg kg^{-1}) (Ma *et al.*, 2018). It is, therefore, considered that *S. suae* could be a good candidate to absorbed As-containing water. While it is also worth noting that *S. suae* is locally found in Taiwan, dominantly available all year, easy to culture and has an established artificial culture (Lin *et al.*, 2018; Lee *et al.*, 2019; Libatique *et al.*, 2020*a*, 2020*b*, 2020*c*) which are important criteria to consider for selecting organisms with potential for bioremediation processes.

In addition, the BCF value of *S. suae* was 96.56 mg kg^{-1} dry weight and 47.96 mg kg^{-1} dry weight of TAs and iAs, respectively (Table 2). These findings demonstrated a higher As capacity than those of Tahira *et al.* (2019) showing a BCF value of only 27 in *E. gracilis*. On the other hand, higher BCF value in other plant and algal species were also reported, for example, *Scenedemus* sp. was reported to have a BCF of 808 and 1014 mg kg^{-1} dry weight of As(III) and As (V) (Bahar *et al.*, 2013), *Wolfia globosa* with BCF value of 940 of As(III) (Zhang *et al.*, 2009) and

Miramthemum umbrosum with higher BCF of 2350 (Islam *et al.*, 2013). Furthermore, macroalgae like *Enteromorpha intestinalis* (green algae) show lower biosorption than the present study with only 4.6 ± 3 and 4.9 mg kg^{-1} of As (Kut *et al.*, 2000; Malea & Kevrekidis, 2014). To compare the biosorption of *S. suae* with other findings, our result has had neither higher nor lower capacity to remove As but we are certain that *S. suae* is an arsenic accumulator and not a hyperaccumulator.

Phosphate effect on total and IAs accumulation by Sarcodia suae

As(V) and phosphate share the same transport pathway in higher plants; the Pi transporter proteins have a higher affinity for phosphate than for As(V) (Meharg *et al.*, 1994). Therefore, As(V) and Pi compete for uptake through the same transport systems in As hyperaccumulators (Tu & Ma, 2003), As-tolerant non-hyperaccumulators (Bleeker *et al.*, 2003), as well as As-sensitive non-accumulators (Esteban *et al.*, 2003). The uptake of As(V) in many organisms (i.e. algae, fungi, plants and amphibians) has been reported to be facilitated by phosphate due to their structural analogy (Gong *et al.*, 2009*a*; Miao *et al.*, 2012; Wang *et al.*, 2013; Zeng *et al.*, 2015). Therefore, phosphate availability or small variations on its concentrations is hypothesized to regulate As(V) uptake in algae (Hellweger *et al.*, 2003) or may significantly change the toxicity of arsenate in algae (Wang *et al.*, 2013).

Sarcodia suae is efficient in absorbing TAs and iAs. Generally, much higher accumulation was pointed out to lower level of phosphate treatments (Fig. 4A–B), an indication of providing good influence of phosphate. Since the structure of a molecule of As (V) is chemically similar to that of the nutrient phosphate, the inability of the phosphate receptors to discriminate against As(V) (Button *et al.*, 1973) and their competitive behaviour for similar transport systems especially when phosphate is present in lower levels may result to higher absorptions. It is also possible that As(V) was taken through phosphate-transporting systems (Zhao *et al.*, 2009) and has been recognized as a phosphate, which is an essential nutrient involved in many biochemical pathways (Goyer & Clarkson, 2001; Hughes, 2002).

Moreover, TAs and iAs uptake were decreased noticeably when the phosphate level was increased. The maximum value was noted in P20 + $As^{5}250$ with 22.95 and 10.26 mg kg^{-1} corresponding amounts of TAs and iAs, respectively (Fig. 4A–B). It can be assumed that *S. suae* can accumulate a high amount of phosphate when available in the environment which was consistent with the statement of Schramm (1991) and thus, phosphate uptake may reduce As. During the As(V) uptake, arsenate might also be obstructed by the luxury uptake of phosphate of the algae which resulted to lower As detected in the algal body. This result could also be confirmed by the reduced phosphate detected in the culture medium. Also, since arsenate can be absorbed with uptake patterns typical of phosphate through the phosphate transport systems, arsenate may be obstructed by phosphate considering its non-specific binding to arsenate reductases (Slaughter *et al.*, 2012). Previous studies show that exposure at high intracellular concentrations of phosphorus may reduce the possibility of As(V) successfully competing with phosphate for enzymatic sites (Meharg *et al.*, 1994), like those studies in *M. aeruginosa* since it is capable of luxury uptake and accumulation of phosphate in several-fold excess of demand when exposed to considerable amounts of phosphate in BG-11 medium before As (V) exposure (Baldia *et al.*, 2007).

Although some studies have established that the phenomenon of arsenic accumulation is not controlled by phosphate concentrations (Knauer & Hemond, 2000; Gong *et al.*, 2009*a*), upon considering such studies in conjunction with our results (Thiel, 1988; Wängberg & Blanck, 1990; Knauer & Hemond, 2000; Gong *et al.*, 2009*a*), we propose that a given phosphate concentration can elicit either positive or negative effects, and this is largely dependent on the arsenic concentration. This phenomenon can be considered from two perspectives. Firstly, if the arsenic concentration is in a suitable or tolerable concentration range, the algae maintain high growth and metabolism, which can allow the algae to resist or metabolize excessive arsenic on its exterior. On the other hand, if the arsenic concentration is too high, it may deliver a dose that is harmful to the algae, thereby inhibiting its growth and related metabolic pathways, and resulting in the regulatory mechanism of arsenic intake no longer being correlated with phosphate concentration. Our previous studies have provided us with enough insight to choose a suitable arsenate dose for this study, and therefore the results of this study were much closer to the former of the two scenarios mentioned above. We believe these two phenomena are helpful in explaining the results of published studies in the current literature and allow us to prove that phosphate and arsenic concentrations in *S. suae* are significantly correlated.

iAs, and total As levels in culture medium

The TAs and iAs levels in the culture medium were dependent on the levels of phosphate. It was noticed that the presence of phosphate in higher levels produced significantly higher

amounts of the As accumulation. On the contrary, significantly lower remaining arsenic was noted in lower phosphate-enriched treatments (Fig. 4C–D).

Furthermore, we also analysed TAs and iAs at 0 and 7 days with no algae of co-existing media (abiotic controls) and results showed that TAs and iAs were detected in the culture medium and similar values were noted even after 7 days of exposure. These findings suggest that *S. suae* accumulated the significant proportions of arsenate in the external medium as reflected by the depletion of arsenate in the culture medium with phosphate and As(V), which unaccounted arsenic was absorbed by the algal body which was confirmed from the results of the intracellular analysis (Fig. 4). While it is also worthy to note that the proportion of iAs remaining in the culture medium was significantly lower in lower phosphate conditions which may be related to the abundance of absorption by the algal body and vice versa. As a result, As ratio was depleted in the culture solution. This result regarding the retention of As by the macroalgae was also attributed to a lower rate of excretion in the external conditions which were in agreement with the findings of Malea & Kevrekidis (2014).

Phosphate depletion in the culture medium

Phosphate depletion was found in the culture medium among treatments (Fig. 5). From these findings, it can be assumed that phosphate was taken up by the algae, phosphate being one of the essential elements for growth (Raghothama, 1999). It is also worthy to note that *S. suae* can accumulate high amounts of phosphate up to 20 mg l^{-1} and this is in agreement with the findings of Schramm (1991) suggesting macroalgae can accumulate a large amount of phosphate from an external medium with high phosphate levels.

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Disclosure statement

No potential conflict of interest was reported by the authors.

Author contributions

Mary Joy Halog Libatique: Methodology, Investigation, Formal analysis, Writing-original draft. Han-Yang Yeh: Methodology, Resources. Fan-Hua Nan: Methodology, Resources. Meng-Chou Lee: Writing-review & editing, Supervision, Funding acquisition. All authors contributed critically to the drafts.

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